Hariom Jangir

Bachelor of Technology
Electrical Engineering
Indian Institute of Technology, Jodhpur

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Portfolio

Education

Bachelor of Technology, Electrical Engineering — *Aug 2024 – Present*
Indian Institute of Technology, Jodhpur — *CGPA: 7.69*

Senior Secondary School — *Apr 2023*
Shree Subhash Bose Senior Secondary School, Ladnun — *Percentage: 94.40%*

Secondary School — *Apr 2021*
Shree Subhash Bose Senior Secondary School, Ladnun — *Percentage: 98%*

Key Courses Taken

Data Structures & Algorithms, Pattern Recognition & Machine Learning, Linear Algebra, Probability, Statistics & Stochastic Processes

Projects

Recipe Recommendation System — *Mar 2025 – Apr 2025*
Tools: Python, Scikit-learn, TF-IDF, PCA, KMeans, KNN [GitHub]

- Engineered an end-to-end ML recommendation pipeline on a dataset of 1,000+ recipes, achieving ~88% cosine similarity accuracy for personalized suggestions.
- Applied TF-IDF vectorization for ingredient text features and reduced dimensionality by 60% using PCA, improving KMeans clustering runtime by 3×.
- Implemented dual-mode querying (recipe-name lookup and custom user query) using cosine similarity and Euclidean distance, enabling flexible recommendation strategies.
- Executed multi-stage feature engineering across text, categorical, and numerical attributes, increasing recommendation relevance by ~25% over a baseline TF-IDF-only model.

Smart Route Finder System — *Jan 2025 – Feb 2025*
Tools: C++, Graph Algorithms, Dijkstra's Algorithm, File I/O [GitHub]

- Architected a graph-based route optimization engine supporting 50+ cities and 200+ weighted edges, computing shortest paths in under 10ms using Dijkstra's algorithm.
- Designed a dynamic CRUD interface for real-time addition, update, and deletion of cities and routes via file-based persistence (`city.txt`, `path.txt`), reducing overhead by 40%.
- Implemented multi-metric route selection (cost, time, and composite score) to enable context-aware path recommendations for diverse travel scenarios.
- Built a multi-city trip planner module using sequential shortest-path computation, supporting 5+ stop itineraries and optimizing total journey distance by up to 30%.

Full-Stack Portfolio Website — *Dec 2024 – Jan 2025*
Tools: React.js, Node.js, Express.js, Three.js, MongoDB, Framer Motion [GitHub]

- Developed a production-grade full-stack portfolio with a React frontend and Express/Node.js REST API backend, deployed on Vercel (frontend) and Render (backend) with zero-downtime CI/CD.
- Integrated an interactive WebGL 3D scene using Three.js and fluid UI transitions with Framer Motion, achieving page load times under 1.5s via code splitting and lazy loading.
- Built a RESTful contact-form API with server-side validation and email notification, handling 100% of form submissions without client-side data loss.
- Designed a fully responsive, mobile-first UI adhering to modern UX principles, achieving a Lighthouse performance score of 90+ across desktop and mobile.

Technical Skills

Languages: C, C++, Python, JavaScript
Frontend: HTML, CSS, Tailwind CSS, React.js, Next.js, Three.js, Framer Motion
Backend: Node.js, Express.js
Database: MongoDB
ML / Data Science: NumPy, Pandas, Scikit-learn, Matplotlib
Tools & Platforms: Git, GitHub, Postman, Google Colab, VS Code, Vercel, Render
Core Concepts: DSA, Object-Oriented Programming (OOP), RESTful API Design
Hardware Description: Verilog